

BOOTSTRAPPING U-STATISTICS

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Summary: Our objective is to review the results on bootstrap theory for U -statistics. The focus is on consistency properties of bootstrap estimators for the distribution function of properly standardized (or studentized) U -statistics. Simple examples are used to explain the reason for bootstrap success or bootstrap failure. General resampling schemes are considered. A survey of material on bootstrapping U -statistics not discussed in the main sections of this paper is included as final section.

1. Introduction

Let $X_1, \dots, X_n$ be a sequence of independent random variables, defined on $(\Omega, \mathcal{B}, \mathbb{P})$, with common distribution function $F(\cdot)$ (an iid sequence). For a statistic $T_n \equiv T_n(X_1, \dots, X_n; F)$ we want to estimate the unknown distribution function $F_{T_n}(x) = \text{Prob}\{T_n \leq x\}$. The bootstrap, introduced by Efron (1979), provides a general tool to construct an estimator for $F_{T_n}(\cdot)$. The main idea is as follows: let $F_n(\cdot)$ be the empirical distribution function, that is, $F_n(x) = n^{-1} \sum_{i=1}^n 1\{X_i \leq x\}$ and, given $X_1, \dots, X_n$, let $X_1^*, \dots, X_n^*$ denote a sequence of independent random variables with common distribution function $F_n(\cdot)$; then the bootstrap estimator for $F_{T_n}(\cdot)$ is defined as

$$F_{T_n^*}(x \mid \mathbf{X}_n) = \text{Pr ob}\{T_n^* \leq x \mid \mathbf{X}_n\}$$

with $T_n^* \equiv T_n(X_1^*, \dots, X_n^*; F_n)$ and $\mathbf{X}_n = (X_1, \dots, X_n)$. For a variety of situations it has been shown that the bootstrap estimator $F_{T_n^*}(\cdot \mid \mathbf{X}_n)$ is a uniformly consistent estimator for $F_{T_n}(\cdot)$, that is, under regularity conditions one can show that

$$\sup_{x \in \mathbb{R}} |F_{T_n^*}(x \mid \mathbf{X}_n) - F_{T_n}(x)| \rightarrow 0, n \rightarrow \infty$$

where the convergence holds in $\mathbb{P}$ -probability (weak bootstrap consistency) or with $\mathbb{P}$ -probability one (strong bootstrap consistency).

One reason for the success of the bootstrap method is that a simple and accurate Monte-Carlo approximation can be given for $F_{T_n^*}(\cdot \mid \mathbf{X}_n)$. The algorithm is as follows:

- (i) For $j = 1, \dots, B$ take independent samples $X_1^*(j), \dots, X_n^*(j)$ from $F_n(\cdot)$ and calculate $T_n^*(j) = T_n(X_1^*(j), \dots, X_n^*(j); F_n)$;
- (ii) Use $B^{-1} \sum_{j=1}^B \mathbf{1}\{T_n^*(j) \leq x\}$ (an approximation of $F_{T_n^*}(x \mid \mathbf{X}_n)$) as estimator for $F_{T_n}(x)$.

Compared to classical asymptotic distribution theory the bootstrap method has some interesting features: it often avoids estimation of complicated asymptotic variance parameters that show up in applications of the central limit theorem; it provides a way to avoid the explicit determination of eigenvalues that appear in limit laws that are of weighted χ^2 -type (see section 2); in a number of situations one can show that the bootstrap estimator is more accurate

than the classical normal approximation (see section 3). On the other hand there are situations where the naive implementation of the algorithm given above leads to inconsistent estimation of $F_{T_n}(\cdot)$ (see section 4).

To get insight into bootstrap success and bootstrap failure we focus attention on one particular class of statistics: U -statistics. The statistical relevance to study this class is explained in Lee (1990): U -statistics play an important role in deriving asymptotic properties of estimators and test statistics since most of them are U -statistics or contain U -statistics as leading components. The paper is organized as follows: in section 2 we give a quick review on asymptotic distribution theory for U -statistics. Bootstrapping non-degenerate U -statistics is discussed in section 3, bootstrap results for degenerate U -statistics are given in section 4. In sections 3 and 4 we restrict attention to Efron's bootstrap. General resampling schemes are discussed in section 5. Finally, in section 6, we give a survey of further results on bootstrapping U -statistics.

2. U -statistics: a quick review

Given a sequence of independent random variables $X_1, \dots, X_n$ with common distribution function F , the study of the sample mean $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ as estimator for the population mean $\mu = EX_1$ is a well established part of probability theory. The notion of averaging over observations $X_1, \dots, X_n$ has been generalized by Hoeffding (1948) in the following way. Consider a measurable function

$$h: \mathbb{R}^m \rightarrow \mathbb{R}: (x_1, \dots, x_m) \mapsto h(x_1, \dots, x_m)$$

symmetric in its m arguments and let $C_{nm} = \{(i_1, \dots, i_m) \in \mathbb{N}^m : 1 \leq i_1 < \dots < i_m \leq n\}$. A U -statistic is obtained by averaging the 'outcomes' $h(X_{i_1}, \dots, X_{i_m}), (i_1, \dots, i_m) \in C_{nm}$, that is,

$$U_n = \binom{n}{m}^{-1} \sum_{C_{nm}} h(X_{i_1}, \dots, X_{i_m}) \quad (2.1)$$

The U -statistic (2.1) is an unbiased estimator of the **regular functional**

$$\begin{aligned} \theta : \mathcal{L}_0 \rightarrow \mathbb{R} : F \mapsto \theta_F &= E h(X_1, \dots, X_m) \\ &= \int_{\mathbb{R}^m} h(x_1, \dots, x_m) dF(x_1) \dots dF(x_m) \end{aligned}$$

where $\mathcal{L}_0$ is a subset of the set $\mathcal{L}$ of one-dimensional distribution functions. The function h is called a **kernel of degree m** . We assume $E|h(X_1, \dots, X_m)| < \infty$. We briefly list some examples.

EXAMPLE 1. Let $\mathcal{L}_0 = \{F \in \mathcal{L} : \int_{\mathbb{R}} x^2 dF(x) < \infty\}$ and let

$\mu = \int_{\mathbb{R}} x dF(x)$. The variance

$$\mu_2 = \int_{\mathbb{R}} (x - \mu)^2 dF(x) = \int_{\mathbb{R}^2} \frac{(x - y)^2}{2} dF(x) dF(y)$$

is a regular functional on $\mathcal{L}_0$. The corresponding U -statistic is

$$U_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} \frac{(X_i - X_j)^2}{2} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2,$$

the sample variance. Note that the minimum variance unbiased estimators of central moments are U -statistics (Hoeffding (1948), p. 295 and Gänssler and Stute (1977), p. 240).

EXAMPLE 2. Let $\mathcal{L}_0 = \{F \in \mathcal{L} : \int_{\mathbb{R}} |x| dF(x) < \infty\}$ and take $h(x, y) = xy$. The corresponding U -statistic

$$U_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} X_i X_j$$

is an unbiased estimator for μ^2 . This simple example will be used later on. ■

EXAMPLE 3. The goodness-of-fit functional

$$\theta_F = \sum_{\ell=1}^k p_{\ell}^{-1} \left(\int_{A_{\ell}} dF - p_{\ell} \right)^2$$

where A_{ℓ} , $\ell = 1, \dots, k$, is a partition of $\mathbb{R}$ into k cells and p_{ℓ} , $\ell = 1, \dots, k$, is a set of specified probabilities attached to the cells, is a regular functional with kernel

$$h(x, y) = \sum_{\ell=1}^k p_{\ell}^{-1} [\mathbf{1}\{x \in A_{\ell}\} - p_{\ell}] [\mathbf{1}\{y \in A_{\ell}\} - p_{\ell}].$$

The corresponding U -statistic is a good approximation for the well-known goodness-of-fit statistic

$$\theta_{F_n} = \sum_{\ell=1}^k p_{\ell}^{-1} (\hat{p}_{\ell n} - p_{\ell})^2$$

where $\hat{p}_{\ell n} = n^{-1} \sum_{i=1}^n \mathbf{1}\{X_i \in A_{\ell}\}$, the empirical version of $\int_{A_{\ell}} dF$. ■

Many more examples can be found in, for example, Lee (1990), chapter 6, Hoeffding (1948), Bickel and Lehmann (1979), Randles and Wolfe (1979), Heilman (1980), de Wet (1987), Janssen *et al.* (1987). Moreover U -statistics are extremely useful to approximate important statistics in for example nonparametric regression estimation and survival analysis (see section 6).

To review the asymptotic distribution theory for U -statistics we take $m = 2$ (to reduce the technicality of the discussion), that is, we consider the U -statistic

$$U_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} h(X_i, X_j).$$

The basic question is to find a **stochastic approximation** $\hat{U}_n$ of iid structure that is close to U_n so that U_n inherits the asymptotic distributional behaviour of $\hat{U}_n$. The appropriate approximation is the projection

$$\hat{U}_n = \sum_{i=1}^n E(U_n | X_i) - (n-1)\theta_F.$$

With $g(x; F) = \int_{\mathbb{R}} h(x, y) dF(y) - \theta_F$, an equivalent form is

$$\hat{U}_n - \theta_F = \frac{2}{n} \sum_{i=1}^n g(X_i; F).$$

If $g \equiv 0$ the U -statistic is called **degenerate** (or **pure**). Otherwise the U -statistic is **non-degenerate**. It is clear that for degenerate U -statistics the projection idea serves no purpose, but that for non-degenerate U -statistics it provides the appropriate tool to find a linear approximation to U_n and hence to establish asymptotic normality. Indeed, with

$$\psi(x, y; F) = h(x, y) - g(x; F) - g(y; F) - \theta_F$$

we have $\sqrt{n}(U_n - \theta_F)/2\sigma_g = S_n + \Delta_n$, where

$$0 < \sigma_g^2 = \text{Var}(g(X_1; F))$$

$$S_n = \frac{\sqrt{n}}{\sigma_g} \sum_{i=1}^n g(X_i; F)$$

and

$$\Delta_n = \frac{\sqrt{n}}{\sigma_g} \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} \psi(X_i, X_j; F)$$

REMARK 1. Note that Δ_n is a degenerate U -statistic.

Assuming $Eh^2(X_1, X_2) < \infty$, Hoeffding (1948) showed that $\Delta_n = o_P(1)$, so that S_n determines the asymptotic distribution of $\sqrt{n}(U_n - \theta_F)/2\sigma_g$.

THEOREM 1. If $Eh^2(X_1, X_2) < \infty$ and $\sigma_g^2 > 0$ (U_n is non-degenerate), then

$$\sqrt{n}(U_n - \theta_F)/2\sigma_g \xrightarrow{d} Z$$

with Z a standard normal random variable. ■

The asymptotic distribution theory for degenerate U -statistics has been developed by Gregory (1977) and Serfling (1980). For $\tilde{h} = h - \theta_F$ define the operator

$$Az(x) = \int_{\mathbb{R}} \tilde{h}(x, y)z(y)dF(y)$$

with $z(\cdot)$ square integrable with respect to F . Let $\lambda_1, \lambda_2, \dots$ denote the (not necessarily distinct) eigenvalues corresponding to the distinct solutions $z_1(\cdot), z_2(\cdot), \dots$ of the equation $Az(\cdot) - \lambda z(\cdot) \equiv 0$.

THEOREM 2. If $Eh^2(X_1, X_2) < \infty$ and $\sigma_g^2 = 0 < \text{Var}(h(X_1, X_2))$ (U_n is degenerate), then

$$n(U_n - \theta_F) \xrightarrow{d} Y$$

with Y a random variable of form $Y = \sum_{j=1}^{\infty} \lambda_j (\chi_j^2(1) - 1)$ where $\chi_1^2(1), \chi_2^2(1), \dots$ are independent $\chi^2(1)$ random variables. ■

EXAMPLE 1 (continued). For the sample variance an application of theorem 1 yields (with μ_k the k -th central moment): if $\mu_4 < \infty$ and $\mu_4 - \mu_2^2 > 0$, then

$$\sqrt{n} \left\{ \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 - \mu_2 \right\}$$

has a limiting normal distribution with mean zero and variance $\mu_4 - \mu_2^2$. ■

EXAMPLE 2 (continued). If $EX_1 = 0$ and $0 < \sigma^2 = EX_1^2 < \infty$, then $U_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} X_i X_j$ is a degenerate U -statistic and $0 < \text{Var}(X_1 X_2) = (EX_1^2)^2 < \infty$. We can therefore apply theorem 2. For this simple example there is only one eigenfunction with $\lambda = \sigma^2$ as corresponding eigenvalue, vid.,

$$x \int_{\mathbb{R}} yz(y) dF(y) = \lambda z(x) \quad (2.2)$$

implies $z(x) = Cx/\lambda$, with $C = \int_{\mathbb{R}} yz(y) dF(y)$. So (2.2) can be written as $Cx\sigma^2/\lambda = Cx$ or equivalently $\lambda = \sigma^2$. We therefore obtain

$$nU_n \xrightarrow{d} \sigma^2(\chi^2(1) - 1). \quad (2.3)$$

Note that for this simple example it is of course possible to prove (2.3) in a direct way. ■

EXAMPLE 3 (continued). Given the null hypothesis it is well known that the asymptotic distribution of $n\theta_{F_n}$, with θ_{F_n} the goodness-of-fit statistic in example 3, is $\chi^2(k-1)$, see, for example, de Wet (1987). An easy calculation yields

$$\theta_{F_n} = \frac{n-1}{n}U_n + \frac{1}{n^2} \sum_{i=1}^n h(X_i, X_i)$$

with $h(x, y)$ and U_n as in example 3. Since

$$\frac{1}{n} \sum_{i=1}^n h(X_i, X_i) \xrightarrow{P} \sum_{\ell=1}^k (1 - p_\ell) \equiv k - 1$$

we have that $nU_n \xrightarrow{d} \sum_{j=1}^{k-1} (\chi_j^2(1) - 1)$ with $\chi_1^2(1), \dots, \chi_{k-1}^2(1)$ independent $\chi^2(1)$ random variables, a result in correspondence with theorem 2.

3. Bootstrapping non-degenerate U -statistics

Given the iid sequence $X_1, \dots, X_n$, let $X_1^*, \dots, X_n^*$ denote a sequence of independent random variables with common distribution $F_n(\cdot)$ (the empirical distribution function). U_n^* , the bootstrap U -statistic for U_n , is obtained simply by replacing the X_i 's by the X_i^* 's. We first show that

$$F_{U_n^*}(x | \mathbf{X}_n) = \text{Prob} \{n^{1/2}(U_n^* - \theta_{F_n}) \leq x | \mathbf{X}_n\} \quad (3.1)$$

is a uniformly consistent estimator for

$$F_{U_n}(x) = \text{Prob} \{n^{1/2}(U_n - \theta_F) \leq x\}.$$

REMARK 2. Note that in (3.1)

$$\theta_{F_n} \equiv E^*(U_n^* | \mathbf{X}_n) = V_n = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n h(X_i, X_j)$$

where $E^*(\cdot | \mathbf{X}_n)$ is the conditional expectation with respect to $\mathbf{X}_n$ (i.e. it denotes the expectation related to the randomness induced by the resampling). If we use V_n for the centering we assume $E | h(X_1, X_1) | < \infty$. We could have used the (asymptotically equivalent) centering $U_n^* - U_n$.

THEOREM 3. If $Eh^2(X_1, X_2) < \infty$, $E | h(X_1, X_1) | < \infty$ and $\sigma_g^2 > 0$, then

$$\sup_{x \in \mathbb{R}} |F_{U_n^*}(x | \mathbf{X}_n) - F_{U_n}(x)| \rightarrow 0, \quad n \rightarrow \infty \text{ a.s. } [\mathbb{P}].$$

REMARK 3. This result is stated as theorem 3.3 in Shao and Tu (1995), p. 81. Bickel and Freedman (1981) were the first to prove this result. Shi (1986) showed that their condition $Eh^2(X_1, X_1) < \infty$ can be relaxed to $E | h(X_1, X_1) | < \infty$. The (new) proof presented here uses the well-known fact (see Efron, 1982) that a bootstrapped statistic can be written in terms of multinomial weights M_{ni} , $i = 1, \dots, n$, and the original sample $X_1, \dots, X_n$. Separating these two sources of randomness emphasizes the double random aspect that is an inherent aspect of bootstrapping. We think that this proof is more transparent than the classical proof.

PROOF OF THEOREM 3. First note that by the strong law of large numbers for U - and V -statistics (see Serfling, 1980, chapters 5 and 6) we have that $\sqrt{n}(U_n - V_n) \rightarrow 0$ a.s. $[\mathbb{P}]$, so that we can work with $U_n^* - U_n$. We use the

following decomposition (see remark 4):

$$U_n^* - U_n = L_n^* + R_n^* \quad (3.3)$$

with

$$L_n^* = \frac{2}{n} \sum_{i=1}^n \{g(X_i^*; F) - g(X_i; F)\}$$

$$R_n^* = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} \{\psi(X_i^*, X_j^*; F) - \psi(X_i, X_j; F)\}.$$

Note that, with $M_{ni} = \#\{j : X_j^* = X_i\}$, we have

$$(M_{n1}, \dots, M_{nn}) \sim \text{Mult} \left(n; \frac{1}{n}, \dots, \frac{1}{n} \right).$$

We can use these multinomial weights to rewrite L_n^* and R_n^* as follows

$$L_n^* = \frac{2}{n} \sum_{i=1}^n (M_{ni} - 1)g(X_i; F)$$

$$R_n^* = R_{n1}^* + R_{n2}^*$$

with

$$R_{n1}^* = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} (M_{ni}M_{nj} - 1)\psi(X_i, X_j; F)$$

and

$$R_{n2}^* = \binom{n}{2}^{-1} \sum_{i=1}^n \frac{M_{ni}(M_{ni} - 1)}{2} \psi(X_i, X_i; F).$$

From Janssen (1994) (or from the classical proof for the bootstrap validity of the sample mean as in, for example, Bickel and Freedman (1981) we have that

$$\sup_{x \in \mathbb{R}} | \text{Prob} \{ \sqrt{n} L_n^* \leq x \mid \mathbf{X}_n \} - \Phi \left(\frac{x}{2\sigma_g} \right) | \rightarrow 0, \quad n \rightarrow \infty \quad \text{a.s. } [\mathbb{P}].$$

It therefore remains to show, for $k = 1, 2$, that

$$\text{Prob} \left\{ n^{1/2} | R_{nk}^* | > \varepsilon \mid \mathbf{X}_n \right\} \rightarrow 0, \quad n \rightarrow \infty \quad \text{a.s. } [\mathbb{P}]. \quad (3.4)$$

We first show that

$$nE^*(R_{n1}^{*2} \mid \mathbf{X}_n) \rightarrow 0, \quad n \rightarrow \infty \quad \text{a.s. } [\mathbb{P}]. \quad (3.5)$$

Note that, with E_M the randomness induced by the multinomial weights,

$$\binom{n}{2}^2 E^*(R_{n1}^{*2} \mid \mathbf{X}_n) \quad (3.6)$$

$$\begin{aligned} &= E_M[(M_{n1}M_{n2} - 1)^2] \sum_{1 \leq i < j \leq n} \psi^2(X_i, X_j; F) \\ &\quad + E_M[(M_{n1}M_{n2} - 1)(M_{n1}M_{n3} - 1)] \sum^{(1)} \psi(X_i, X_j; F) \psi(X_k, X_\ell; F) \\ &\quad + E_M[(M_{n1}M_{n2} - 1)(M_{n3}M_{n4} - 1)] \sum^{(0)} \psi(X_i, X_j; F) \psi(X_k, X_\ell; F) \end{aligned}$$

where $\sum^{(1)}$, resp. $\sum^{(0)}$, denotes summation over $i < j$, $k < \ell$ and $\#\{i, j\} \cap \{k, \ell\} = 1$, resp. 0 (one index, resp. zero indices, in common).

Based on the explicit expressions for mixed factorial moments of $(M_{n1}, \dots, M_{nn})$ (see e.g. Johnson and Kotz, 1969, p. 284) we easily obtain

$$E_M[(M_{n1}M_{n2} - 1)^2] = 3 + O(n^{-1}) \quad (3.7)$$

$$E_M[(M_{n1}M_{n2} - 1)(M_{n1}M_{n3} - 1)] = 1 + O(n^{-1}) \quad (3.8)$$

$$E_M[(M_{n1}M_{n2} - 1)(M_{n3}M_{n4} - 1)] = O(n^{-1}). \quad (3.9)$$

We further have, by the SLLN for U -statistics, that

$$\binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} \psi^2(X_i, X_j; F) \rightarrow E\psi^2(X_1, X_2; F), \quad n \rightarrow \infty \text{ a.s. } [\mathbb{P}] \quad (3.10)$$

$$\binom{n}{3}^{-1} \sum^{(1)} \psi(X_i, X_j; F) \psi(X_k, X_\ell; F) \rightarrow 0, \quad n \rightarrow \infty \text{ a.s. } [\mathbb{P}] \quad (3.11)$$

$$\binom{n}{4}^{-1} \sum^{(0)} \psi(X_i, X_j; F) \psi(X_k, X_\ell; F) \rightarrow 0, \quad n \rightarrow \infty \text{ a.s. } [\mathbb{P}]. \quad (3.12)$$

From (3.6)–(3.12) we easily obtain (3.5) and hence (3.4) for $k = 1$. For $k = 2$ note that

$$\begin{aligned} & n^{1/2} \binom{n}{2}^{-1} E^*[\|R_{n2}^* \| \mathbf{X}_n] \\ & \leq n^{1/2} \binom{n}{2}^{-1} \frac{1}{2} E_M[M_{n1}(M_{n1} - 1)] \sum_{i=1}^n |\psi(X_i, X_i)| \\ & = n^{-1/2} \frac{1}{n} \sum_{i=1}^n |\psi(X_i, X_i)| \rightarrow 0, \quad n \rightarrow \infty \text{ a.s. } [\mathbb{P}]. \end{aligned}$$

This completes the proof. ■

REMARK 4. Note that the decomposition for U_n^* used in the previous proof is the appropriate one if one thinks of bootstrapping as ‘reweighting the outcomes $h(X_i, X_j)$, $1 \leq i < j \leq n$, by multinomial weights’. If one thinks of bootstrapping as ‘taking a sample from the empirical distribution function

$F_n(\cdot)$ one rather will decompose U_n^* in terms of $g(x; F_n)$ and $\psi(x, y; F_n)$. See, for example, Helmers (1991).

REMARK 5. Theorem 3 immediately suggests two further problems: give a standardized and studentized version of theorem 3 and provide information on the uniform consistency rate. Answers to these problems are given by Shi (1987) and Helmers (1991). With

$$S_n^2 = \frac{4(n-1)}{(n-2)^2} \sum_{i=1}^n \left(\frac{1}{n-1} \sum_{j=1}^n h(X_i, X_j) - U_n \right)^2,$$

the jackknife estimator for $n \text{Var}(U_n)$ (see Callaert and Veraberbeke, 1981), we have the following results.

THEOREM 4. (Shi, 1987). If $E | h^3(X_1, X_2) | < \infty$, $E | h(X_1, X_1) | < \infty$ and $\sigma_g^2 > 0$, then

$$\begin{aligned} \sup_{x \in \mathbb{R}} & \left| \text{Prob} \left\{ \frac{\sqrt{n}(U_n^* - V_n)}{S_n} \leq x \mid \mathbf{X}_n \right\} \right. \\ & \left. - \text{Prob} \left\{ \frac{\sqrt{n}(U_n - \theta_F)}{2\sigma_g} \leq x \right\} \right| \\ & = O(n^{-1/2}) \qquad \text{a.s. } [\mathbb{P}]. \end{aligned}$$

THEOREM 5. (Helmers, 1991). If $E | h(X_1, X_2) |^{4+\varepsilon} < \infty$, for some $\varepsilon > 0$, $E | h^3(X_1, X_1) | < \infty$ and the distribution function of $g(X_1; F)$ is nonlattice, then

$$\sup_{x \in \mathbb{R}} \left| \text{Prob} \left\{ \frac{\sqrt{n}(U_n^* - V_n)}{S_n^*} \leq x \mid \mathbf{X}_n \right\} - \text{Prob} \left\{ \frac{\sqrt{n}(U_n - \theta_F)}{S_n} \leq x \right\} \right|$$

$$= o(n^{-1/2}) \quad \text{a.s. } [\mathbb{P}] \quad \blacksquare$$

In theorem 5 S_n^* is obtained from S_n by replacing the X_i 's by X_i^* 's. Theorem 5 shows that the bootstrap approximation is asymptotically closer to the exact distribution of a studentized U -statistic than the normal approximation, for which the Berry-Esseen rate $O(n^{-1/2})$ is valid (Callaert and Veraverbeke, 1981; Helmers, 1985). As an application, improved bootstrap-based confidence intervals for the mean of a U -statistic can be obtained. See Helmers (1991) for a thorough discussion.

We finally mention that in Tu (1992) transformation-based bootstrap approximations for the exact distribution of U -statistics are studied. They turn out to be more accurate than the normal approximations.

4. Bootstrapping degenerate U -statistics

Based on example 2 we show that the naive bootstrap does not work for degenerate U -statistics. Then we show that recentering of the kernel is needed to get the appropriate bootstrap version for a given degenerate U -statistic.

EXAMPLE 2 (continued). From section 2 we know that, if $EX_1 = 0$ and $0 < \sigma^2 = EX_1^2 < \infty$,

$$nU_n \xrightarrow{d} \sigma^2(\chi^2(1) - 1). \quad (4.1)$$

Since $V_n = \bar{X}_n^2$ and $U_n^* = \frac{n}{n-1} \left(\bar{X}_n^{*2} - \frac{1}{n^2} \sum_{i=1}^n X_i^{*2} \right) \equiv \frac{n}{n-1} \left(V_n^* - \frac{1}{n^2} \sum_{i=1}^n X_i^{*2} \right)$ we have, up to a multiplicative constant tending

to one,

$$n(U_n^* - V_n) = n(\bar{X}_n^{*2} - \bar{X}_n^2) - n^{-1} \sum_{i=1}^n X_i^{*2}.$$

By the bootstrap law of large numbers (see section 6(a)) we have that, except for the centering of σ^2 in (4.1), $\text{Prob} \{n(U_n^* - V_n) \leq x \mid \mathbf{X}_n\}$ asymptotically behaves like $\text{Prob} \{n(V_n^* - V_n) \leq x \mid \mathbf{X}_n\}$.

Now follow Babu (1984): if $\bar{X}_n > 0$ we have for any $x \geq 0$

$$\begin{aligned} \text{Prob} \{n(V_n^* - V_n) \leq x \mid \mathbf{X}_n\} & \quad (4.2) \\ & \geq \text{Prob} \{n(V_n^* - V_n) \leq 0 \mid \mathbf{X}_n\} \\ & = \text{Prob} \{-\bar{X}_n \leq \bar{X}_n^* \leq \bar{X}_n \mid \mathbf{X}_n\} \\ & = \text{Prob} \{-2n^{1/2} \bar{X}_n \leq \sqrt{n}(\bar{X}_n^* - \bar{X}_n) \leq 0 \mid \mathbf{X}_n\} \\ & \rightarrow \Phi(0) = \frac{1}{2}. \end{aligned}$$

Since $nV_n \xrightarrow{d} \sigma^2 \chi^2(1)$, (4.2) demonstrates that $\text{Prob} \{n(V_n^* - V_n) \leq x \mid \mathbf{X}_n\}$ is an inconsistent estimator for $\text{Prob} \{nV_n \leq x\}$ (and $\text{Prob} \{n(U_n^* - V_n) \leq x \mid \mathbf{X}_n\}$ is inconsistent as estimator for $\text{Prob} \{nU_n \leq x\}$).

A different way to see this is as follows (Shao, 1994):

$$n(\bar{X}_n^{*2} - \bar{X}_n^2) = 2n^{1/2} \bar{X}_n \sqrt{n}(\bar{X}_n^* - \bar{X}_n) + n(\bar{X}_n^* - \bar{X}_n)^2. \quad (4.3)$$

Since $\limsup_{n \rightarrow \infty} n^{1/2} \bar{X}_n / \sqrt{\log \log n} = \sigma$ a.s. [P] (the LIL), the first term in the r.h.s. of (4.3) does not have a weak limit.

The deeper mathematical reason for this problem is that, for degenerate U -statistics in general, $n(U_n - \theta_F)$ is not uniformly continuous as function of F , a fact already observed by Bretagnolle (1983), see also Dehling and Mikosch (1994), p. 394. ■

REMARK 6. The following observation due to van Zwet (see Mammen, 1992, p. 37) provides a different way to see that naive bootstrapping does not work for degenerate U -statistics: if $Eh^2(X_1, X_1) < \infty$, $Eh^2(X_1, X_2) < \infty$ and $g(x; F) = 0$, then

$$n^2[E\text{Var}^*(U_n^* | \mathbf{X}_n) - 3\text{Var}(U_n)] \rightarrow 0.$$

To remedy this inconsistency centering of the kernel is needed. To understand this it is instructive to first look at the sample mean as a tutorial example (see Dehling and Mikosch, 1994, p. 394–395).

EXAMPLE 4. Again we look at an iid sequence $X_1, \dots, X_n$ with $EX_1 = 0$ and $0 < \sigma^2 = EX_1^2 < \infty$. We then have $\text{Prob}\{n^{1/2}\bar{X}_n \leq x\} \rightarrow \Phi(x/\sigma)$. The naive bootstrap estimator is $\text{Prob}\{n^{1/2}\bar{X}_n^* \leq x | \mathbf{X}_n\}$ and this can be rewritten as $\text{Prob}\{\sqrt{n}(\bar{X}_n^* - \bar{X}_n) + n^{1/2}\bar{X}_n \leq x | \mathbf{X}_n\}$ and by the law of iterated logarithm, the estimator is not strongly consistent.

Since, with $\mu = EX_1 = 0$, $n^{1/2}\bar{X}_n = \sqrt{n}(\bar{X}_n - \mu)$ the correct bootstrap version is $\sqrt{n}(\bar{X}_n^* - \bar{X}_n)$. This shows that in the estimator given above the term $n^{1/2}\bar{X}_n$ is abundant. Another way to say this is that we have to resample

from the centered X_i 's, that is, we resample from $X_1 - \bar{X}_n, \dots, X_n - \bar{X}_n$ (location alignment) since then we have $E^*(X_i^* | \mathbf{X}_n) = 0$. This means that now the bootstrap world (the X_i^* 's) and the real world (the X_i 's) (see Efron and Tibshirani, 1993, p. 87 for this terminology) share the zero-mean property, that is, we work under the same model assumption in both worlds. This is crucial to get a consistent bootstrap procedure. The simple example shows that the values used in the resampling are determined by the underlying model. It is very important to realize that underlying model assumptions will provide guidance on how to set up resampling (see Shao and Tu, 1995, p. 17). ■

A similar problem shows up with degenerate U -statistics. In the real world we have that $g(x; F) = 0$ but in the bootstrap world we have that $g(x; F_n) \neq 0$, so that the bootstrapped U -statistic U_n^* does have a linear term. Therefore instead of normalizing with n we should have normalized with $\sqrt{n}$, hence the bootstrap inconsistency. Based on example 4 we know that the appropriate remedy is to mimic in the bootstrap world what is going on in the real world. We therefore define a kernel that is degenerate in the bootstrap world (note that $V_n = \theta_{F_n}$)

$$\begin{aligned}\bar{h}_n(x, y; F_n) &= h(x, y) - g(x; F_n) - g(y; F_n) - \theta_{F_n} \\ &= h(x, y) - \frac{1}{n} \sum_{k=1}^n h(x, X_k) - \frac{1}{n} \sum_{k=1}^n h(X_k, y) + V_n\end{aligned}\quad (4.4)$$

(see (2.24) in Arcones and Giné, 1992).

The corresponding U -statistic

$$U_n^*(\bar{h}_n, F_n) = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} \bar{h}(X_i^*, X_j^*; F_n)$$

is the appropriate bootstrap estimator for U_n . We indeed have the following theorem.

THEOREM 6. If $Eh^2(X_1, X_2) < \infty$, $E | h(X_1, X_1) | < \infty$ and $g(x; F) = 0$, then

$$\sup_{x \in \mathbb{R}} | \text{Prob} \{ nU_n^*(\bar{h}_n, F_n) \leq x \mid \mathbf{X}_n \} - \text{Prob} \{ n(U_n - \theta_F) \leq x \} | \rightarrow 0, \quad n \rightarrow \infty \quad \text{a.s. } [\mathbb{P}].$$

REMARK 7. Theorem 6 is due to Arcones and Giné (1992). A different and simple way to prove this result is given in Dehling and Mikosch (1994). They show that the sample distribution and the bootstrap distribution are close in a modified Mallows metric, a result that implies bootstrap consistency.

REMARK 8. Using an appropriate version of the bootstrap law of large numbers (see section 6(a)) one can show that

$$U_n^*(\bar{h}_n, F_n) = \sum_{i \neq j} \left(\frac{M_{ni}}{n} - \frac{1}{n} \right) \left(\frac{M_{nj}}{n} - \frac{1}{n} \right) h(X_i, X_j) + o_{P^*}(n^{-1})$$

a.s. $[\mathbb{P}]$.

This expression clearly shows that we recenter the weights so that we 'create' degeneracy in the bootstrap world.

5. U -statistics and weighted bootstrapping

From the proof of theorem 3 it is immediate that, for nondegenerate U -statistics, we could have defined the bootstrap version of U_n by

$$U_{M_n} = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} M_{ni} M_{nj} h(X_i, X_j) \approx \sum_{i \neq j} \frac{M_{ni}}{n} \frac{M_{nj}}{n} h(X_i, X_j). \quad (5.1)$$

Note that $U_{M_n} = U_n^* + o_{P^*}(n^{-1/2})$ a.s. $[\mathbb{P}]$ (with P^* a shorthand notation for the (conditional) bootstrap probability).

From remark 8 we know that the appropriate bootstrap version for a degenerate U -statistic U_n is given by

$$U_{M_n}^c = \sum_{i \neq j} \left(\frac{M_{ni}}{n} - \frac{1}{n} \right) \left(\frac{M_{nj}}{n} - \frac{1}{n} \right) h(X_i, X_j). \quad (5.2)$$

From (5.1) and (5.2) it is clear that Efron's bootstrap is (essentially) obtained by assigning random (multinomial) weights (or random coefficients) to the outcomes $h(X_i, X_j)$, $1 \leq i < j \leq n$. So the natural question arises whether other weighting schemes (i.e. other resampling plans) lead to consistent bootstrap estimators for the distribution function of the U -statistic. This problem, called **weighted bootstrapping** or **generalized bootstrapping**, has been studied by Mason and Newton (1992) for the sample mean. Weighted bootstrap for U -statistics has been considered by Janssen (1994), Hušková and Janssen (1993a, 1993b) and Dehling and Mikosch (1994). A further relevant reference is Barbe and Bertail (1995). We briefly summarize the main results.

Let $W_n = (W_{n1}, \dots, W_{nn})$ be a vector of random weights defined on a probability space $(\tilde{\Omega}, \tilde{B}, \tilde{\mathbb{P}})$ and independent of the data $X_1, \dots, X_n$. We assume that the elements of W_n are exchangeable and that

$$(W.1) \quad W_{ni} \geq 0, \quad i = 1, \dots, n \quad \text{and} \quad \sum_{i=1}^n W_{ni} = 1;$$

$$(W.2) \quad n \sum_{i=1}^n \left(W_{ni} - \frac{1}{n} \right)^2 \xrightarrow{\tilde{\mathbb{P}}} c^2, \quad n \rightarrow \infty, \quad \text{for some } c > 0;$$

$$(W.3) \quad n \max_{1 \leq i \leq n} \left(W_{ni} - \frac{1}{n} \right)^2 \xrightarrow{\tilde{\mathbb{P}}} 0, \quad n \rightarrow \infty.$$

In terms of $\mathbf{W}_n$ the generalized bootstrap analogues of U_{M_n} , $U_{M_n}^c$ and S_n^{*2} (as defined in theorem 5) are

$$U_{\mathbf{W}_n} = \sum_{i \neq j} W_{ni} W_{nj} h(X_i, X_j)$$

$$U_{\mathbf{W}_n}^c = \sum_{i \neq j} \left(W_{ni} - \frac{1}{n} \right) \left(W_{nj} - \frac{1}{n} \right) h(X_i, X_j)$$

and

$$S_{\mathbf{W}_n}^2 = 4 \sum_{i=1}^n W_{ni} \left(P_{\mathbf{W}_n}^{(i)} - V_{\mathbf{W}_n} \right)^2$$

where

$$P_{\mathbf{W}_n}^{(i)} = \sum_{j=1}^n W_{nj} h(X_i, X_j) \quad \text{and} \quad V_{\mathbf{W}_n} = \sum_{i=1}^n \sum_{j=1}^n W_{ni} W_{nj} h(X_i, X_j).$$

A direct calculation yields

$$S_{\mathbf{W}_n}^2 = 4 \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n W_{ni} W_{nj} W_{nk} h(X_i, X_j) h(X_i, X_k) - 4V_{\mathbf{W}_n}^2.$$

For non-degenerate U -statistics we have the following theorem where $F_{U_n}(x)$ is defined by (3.2) and with

$$G_{U_n}(x) = \text{Prob} \{ \sqrt{n}(U_n - \theta_F)/S_n \leq x \};$$

$$F_{U_{\mathbf{W}_n}}(x) = \text{Prob} \{ \sqrt{nc}^{-1}(U_{\mathbf{W}_n} - U_n) \leq x \mid \mathbf{X}_n \};$$

$$G_{U_{\mathbf{W}_n}}(x) = \text{Prob} \{ \sqrt{nc}^{-1}(U_{\mathbf{W}_n} - U_n)/S_{\mathbf{W}_n} \leq x \mid \mathbf{X}_n \}.$$

THEOREM 7. (Hušková and Janssen, 1993a). If $Eh^2(X_1, X_2) < \infty$ and $\sigma_g^2 > 0$, then

$$\sup_{x \in \mathbb{R}} |F_{U_{W_n}}(x | \mathbf{X}_n) - F_{U_n}(x)| \rightarrow 0, \quad n \rightarrow \infty \quad \text{a.s.} \quad [\mathbb{P}].$$

Adding the condition $E|h(X_1, X_1)| < \infty$, we have

$$\sup_{x \in \mathbb{R}} |G_{U_{W_n}}(x | \mathbf{X}_n) - G_{U_n}(x)| \rightarrow 0, \quad n \rightarrow \infty \quad \text{a.s.} \quad [\mathbb{P}]. \quad \blacksquare$$

For degenerate U -statistics we have the following result.

THEOREM 8. (Hušková and Janssen, 1993b). If $Eh^2(X_1, X_2) < \infty$ and $\sigma_g^2 = 0$, then

$$\sup_{x \in \mathbb{R}} |\text{Prob}\{nU_{W_n}^c \leq x | \mathbf{X}_n\} - \text{Prob}\{n(U_n - \theta_F) \leq x\}| \rightarrow 0, \quad n \rightarrow \infty$$

a.s. $[\mathbb{P}]$. \blacksquare

Note that multinomial weights are exchangeable weights satisfying (W.1) – (W.3), that is, Efron's resampling scheme is a special case of weighted bootstrapping. Another choice that is well known by now are weights of form $W_{ni} = \zeta_i/n\bar{\zeta}$ with $\zeta_1, \dots, \zeta_n$ any sequence of strictly positive iid random variables (with mean μ and finite variance σ^2 and (in W_2) $c^2 = \sigma^2/\mu^2$). If we take the ζ_i 's exponential with $\mu = 1$ we obtain the Dirichlet weights (see the papers on Bayesian bootstrap by Rubin (1981) and Lo (1987) for further details). See Hušková and Janssen (1993b) for more details on these examples. Interesting further examples, including deterministic jackknife weights, are considered in Praestgaard and Wellner (1993) and Hall and Mammen (1994).

REMARK 9. From theorems 8 and 9 two immediate questions arise:

- (i) why consider general weights; and
- (ii) how to choose the weights.

Answer to the why-question: The approach used to establish the uniform consistency result for the bootstrap distribution function is based on rank statistic methodology which provides a unified procedure capable of handling different resampling schemes. Considering general weights provides the key to compare different resampling schemes. Since all schemes are first order equivalent, such comparison will require higher order asymptotics. See also Shao and Tu (1995), chapter 10 for a small simulation study.

Answer to the how-question: Barbe and Bertail (1995) devote a special chapter to the choice of the weights. They write: "It would be extremely surprising that all weights with $EW_{ni} = 1/n$ lead to the same result in practice" (Barbe and Bertail, 1995, p. 45).

The main part in their work is to show that the assumption of $\mathbf{W}_n$ being independent of $X_1, \dots, X_n$ can be dropped. As a result the weights can be chosen as a function of the data. For such data-dependent weights they show that bootstrap critical points can be chosen so that the prescribed coverage probability is obtained at a faster rate than what one gets for example from Efron's bootstrap (Barbe and Bertail, 1995, p. 52–53); they further illustrate that the bootstrap distribution can be matched with Edgeworth expansions up to $o_P(n^{-1/2})$ (Barbe and Bertail, 1995, p. 58). From the reading of their chapter 2 it is clear that many issues about choosing the weights are still unsolved.

However, one clear message is already available: the optimal choice of the weights is model dependent. Further relevant information and motivation on the weighted bootstrap can be obtained from Häusler *et al.* (1992) and Shao and Tu (1995), chapter 10. The latter reference also gives a nice introductory discussion on the Bayesian bootstrap (i.e. Dirichlet weights).

6. Further results on bootstrapping U -statistics

In the previous sections we discussed consistency results for bootstrap estimators of the distribution function of a U -statistic (non-degenerate or degenerate). Much more material and further extensions are of course available in the statistical literature. The idea of this final section is to briefly guide the reader through this material.

(a) *The bootstrap of the law of large numbers for U -statistics*

Arcones and Giné (1992) prove the following result for the Efron bootstrap.

THEOREM 9. If $E | h(X_1, X_2) | < \infty$ and $E | h(X_1, X_1) |^{1/2} < \infty$, then

$$U_n^* \xrightarrow{P^*} Eh(X_1, X_2) \quad \text{a.s. } [P].$$

REMARK 10. Arcones and Giné (1992), p. 670 established a bootstrap law of large numbers for U -statistic with kernel of arbitrary degree m . If we specialize this result to the sample mean, we get:

if $E | X_1 | < \infty$, then $\bar{X}_n^* \xrightarrow{P^*} \mu = EX_1 \quad \text{a.s. } [P]$; or

if $E |X_1| < \infty$, then $\bar{X}_n^* - X_n \xrightarrow{P^*} 0$ a.s. $[\mathbb{P}]$.

Relevant references for further reading are Athreya *et al.* (1984), Csörgő (1992) and Mikosch (1994).

(b) *Bootstrapping U-statistics with estimated parameters*

Resampling methods for U -statistics with kernels containing an estimated parameter are studied in Janssen and Veraverbeke (1992) and Arcones and Giné (1994). In both papers consistency of the bootstrap estimator for the distribution function of such U -statistics is shown. These results apply to U -statistics based on trimmed samples and U -statistics based on aligned samples.

(c) *Bootstrapping k-sample U-statistics*

The validity of bootstrap approximations for the distribution of a (standardized/studentized) k -sample statistic has been studied in a number of situations (see e.g. Hall and Martin, 1988). In Boos *et al.* (1989) bootstrap estimation of the distribution of a two-sample U -statistic is studied when resampling is from residuals (centered data or location-aligned data). Based on these results bootstrap critical points for homogeneity tests are obtained. In that paper it is also explained that naive bootstrapping leads to unacceptable testing procedures. A further relevant reference is Boos and Brownie (1989).

(d) *Bootstrapping U-quantiles*

Let $H_F(y) = \text{Prob} \{h(X_1, \dots, X_m) \leq y\}$, $y \in \mathbb{R}$, denote the distribution

function of the random variable $h(X_1, \dots, X_m)$ and let

$$H_n(y) = \binom{n}{m}^{-1} \sum_{C_{nm}} \mathbf{1}\{h(X_{i_1}, \dots, X_{i_m}) \leq y\} \quad (6.1)$$

denote the empirical counterpart, the empirical distribution function of U -statistic structure (since for every fixed y (6.1) is a U -statistic (see (2.1))). For $0 < p < 1$ let $\xi_p = H_F^{-1}(p)$ and $\hat{\xi}_{pn} = H_n^{-1}(p)$. Let $H_n^*(\cdot)$ denote the empirical distribution function of U -statistic structure based on a bootstrap sample $X_1^*, \dots, X_n^*$ from $F_n(\cdot)$ and let $\hat{\xi}_{pn}^* = H_n^{*-1}(p)$, for $0 < p < 1$. Helmers *et al.* (1992) proved the following.

THEOREM 10. If $H_F(\cdot)$ is continuously differentiable (with density $h_F(\cdot)$) in a neighbourhood of ξ_p with $h_F(\xi_p) > 0$, then

$$\sup_{x \in \mathbb{R}} |\text{Prob}\{\sqrt{n}(\hat{\xi}_{pn}^* - \hat{\xi}_{pn}) \leq x \mid X_n\}| \quad (6.2)$$

$$- \text{Prob}\{\sqrt{n}(\hat{\xi}_{pn} - \xi_p) \leq x\} \rightarrow 0, \quad n \rightarrow \infty$$

a.s. $[\mathbb{P}]$.

They also show that the rate of convergence in (6.2) is $O(n^{-1/4}(\log n)^{3/4})$ a.s. $[\mathbb{P}]$. Helmers and Hušková (1994) extend this result to multivariate U -quantiles. Arcones (1995) shows that, if $g(x, t) = \text{Prob}\{h(x, Y) \leq t\}$ is smooth, the rate in (6.2) can be improved to $O(n^{-1/2}(\log \log n)^{1/2})$ a.s. $[\mathbb{P}]$. Finally note that for $m = 2$ Aerts and Janssen (1995) show that (6.2)

remains valid if $\hat{\xi}_{pn}^*$ is replaced by $\hat{\xi}_{\mathbf{W}_n, p} = H_{\mathbf{W}_n}^{-1}(p)$ with

$$H_{\mathbf{W}_n}(y) = \sum_{i \neq j} W_{ni} W_{nj} \mathbf{1}\{h(X_i, X_j) \leq y\} / \left(1 - \sum_{i=1}^n W_{ni}^2\right)$$

where the weights satisfy (W.1) – (W.3) in section 5.

(e) *U-statistics and nonparametric regression*

Mammen (1992), p. 36, shows that a certain goodness-of-fit test (measuring the distance between a parametric and a nonparametric regression estimate) can be represented (up to lower order terms) as a degenerate U -statistic. For this example he shows (as we could expect from section 4) that the naive bootstrap and the adjusted residuals bootstrap (Mammen, 1992, p. 35) lead to inconsistency. He further shows that the wild bootstrap leads to consistency. This example once more illustrates that, according to the underlying model, only specific resampling schemes lead to consistency.

(f) *U-statistics and survival analysis*

In survival analysis, as in nonparametric regression, U -statistics are often good approximations for statistics of interest. Therefore showing the validity of bootstrap approximation will rely heavily on appropriate results for U -statistics (see Veraverbeke, 1997, for examples and precise references).

(g) *Estimating standard errors*

So far we restricted attention to bootstrap estimation of (unknown) distribution functions. We should realize, however, that the problem of estimating the standard error of a given statistic initiated the study of resampling meth-

ods (bootstrap method and of course also the jackknife method). Therefore within the U -statistic context a relevant track is to use bootstrapping and jackknifing to estimate the standard error of a given U -statistic. The problem is discussed in detail in Lee (1990), chapter 5. Here we also mention Efron and Stein (1981) as important reference.

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